

Physics 140 – CSP section

Vectors

Lecture 01

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1484(A) Randall Lab

Scalars and Vectors

SCALAR - quantity that can be described by a single number:

e.g. temperature, mass, amount of money.

VECTOR - quantity having both magnitude and direction:

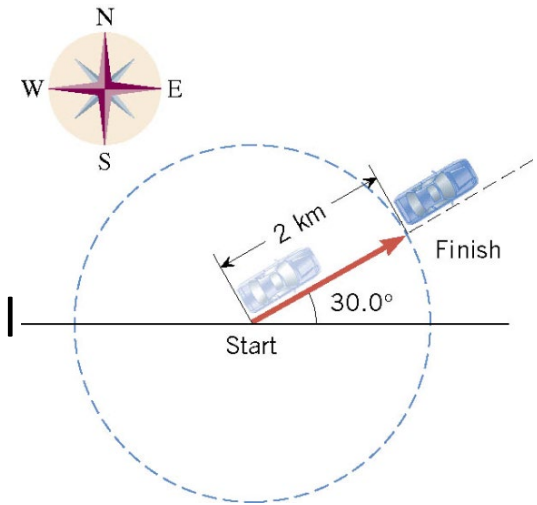
e.g. displacement, velocity, acceleration, force, momentum.

- The magnitude always has a positive value – a number.
- The direction is specified mathematically by an angle which can be positive or negative – another number.

Geometric Representation of Vectors

A Vector is represented (geometrically) by an Arrow.

- Magnitude of Vector = Length of Arrow
[e.g. magnitude of (vector) displacement = 2 km = 2000 m]
- Direction of Vector = Angle of Arrow with horizontal



In Mathematics:

Direction = Angle the Vector (Arrow) makes with a conventionally chosen directed line (axis). – e.g. Here: **30° North of East.**

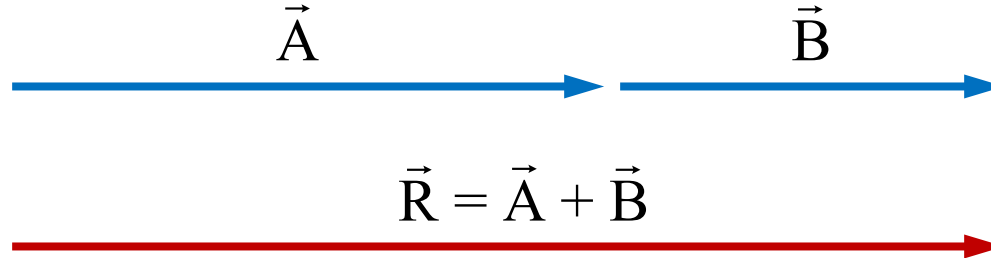
30° North of East = 30° to the North starting from the East

Important NOTE:

Except for the particular case of vectors along the same line [called one-dimensional (1D) vectors], *most* vectors DO NOT add or subtract like scalars ! There is a GEOMETRY *issue* at play which HAS to be taken into consideration every time we are dealing with vectors *in plane* [called two-dimensional (2D) vectors] or *in space* [called three-dimensional (3D) vectors].

This is shown in the next slides for 1D and 2D vectors.

Addition of Two Vectors Along a Line (1D)



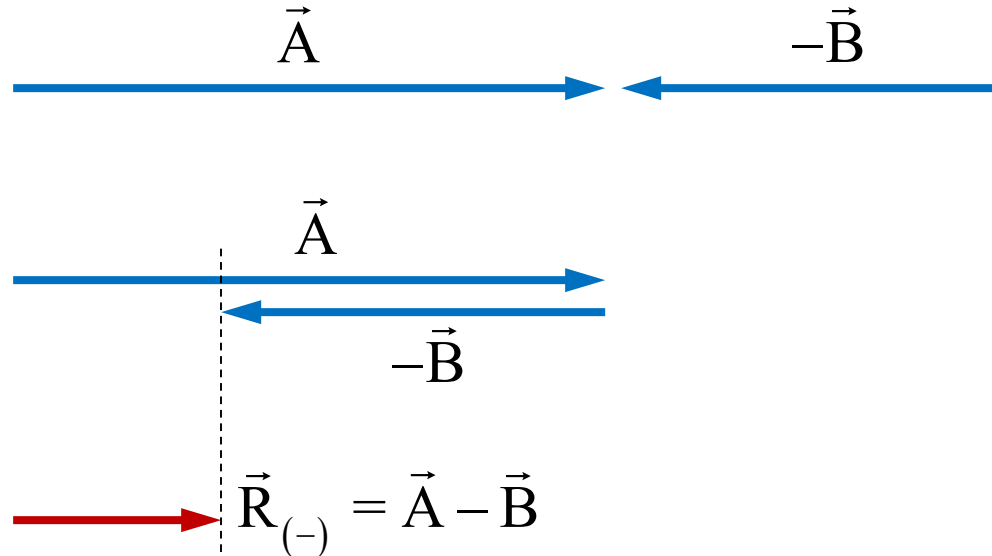
$\vec{A}$ = vector A, $\vec{B}$ = vector B, $\vec{R}_{(+)}$ = resultant *sum vector*

$\vec{R}_{(+)} = \vec{A} + \vec{B} = \vec{B} + \vec{A}$, where $\vec{B} \parallel \vec{A}$ and in the *same* direction.

MAGNITUDE of resultant *sum vector* $\vec{R}_{(+)}$:

$$R_{(+)} = \left| \vec{R}_{(+)} \right| = A + B$$

Subtraction of Two Vectors Along a Line (1D)

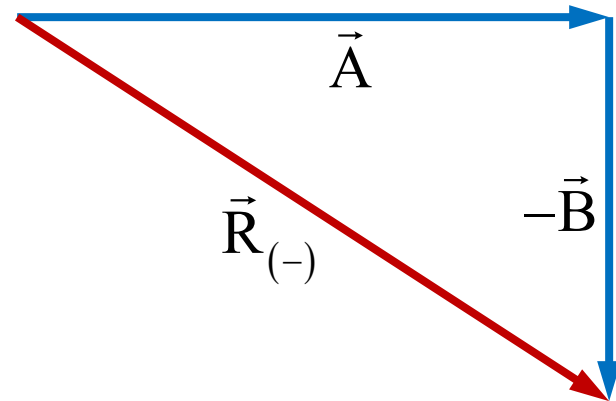
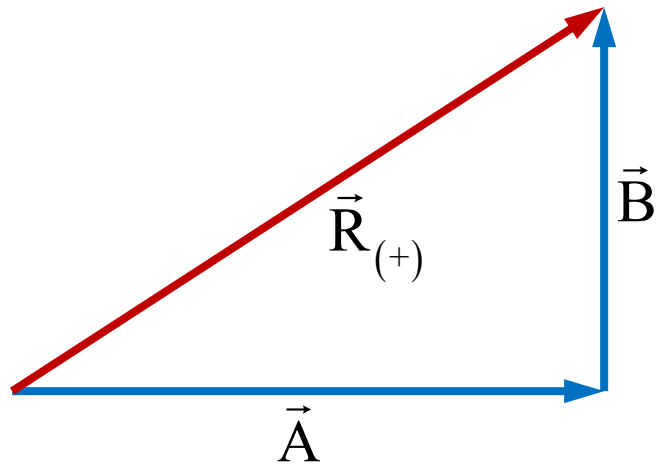


$$\vec{R}_{(-)} = \vec{A} - \vec{B} = \vec{A} + (-\vec{B}) = (-\vec{B}) + \vec{A}, \text{ where } \vec{B} \parallel \vec{A} \text{ but in } \textit{opposite} \text{ direction}$$

MAGNITUDE of resultant *difference vector* $\vec{R}_{(-)}$:

$$R_{(-)} = \left| \vec{R}_{(-)} \right| = |A - B|$$

Addition of Two Perpendicular Vectors (2D)



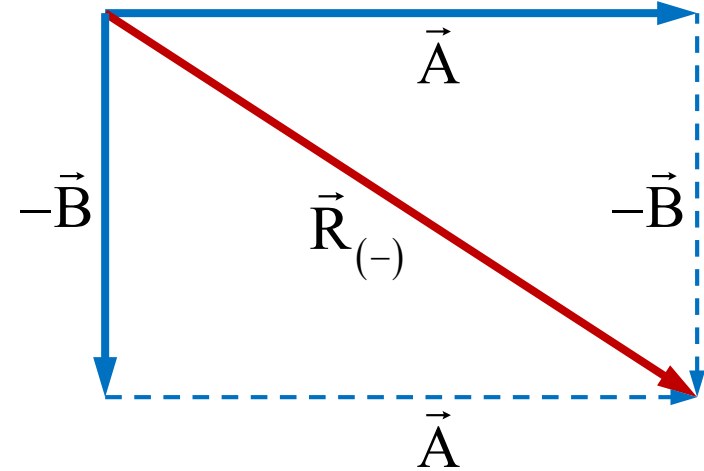
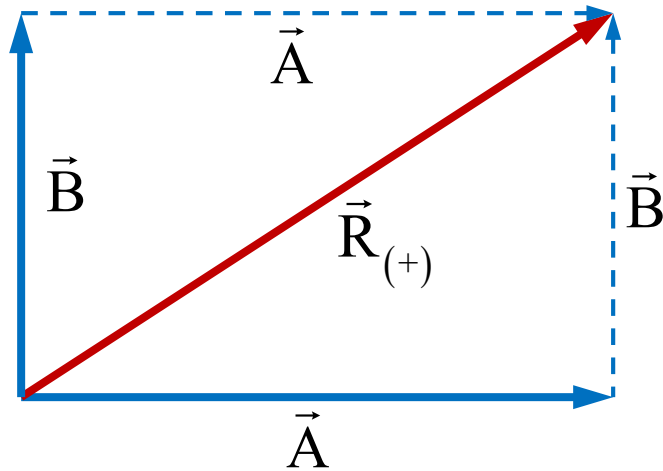
Polygon Method (Tip - to - Toe / Head - to - Tail)

$$\vec{R}_{(+)} = \vec{A} + \vec{B} \quad \text{and} \quad \vec{R}_{(-)} = \vec{A} - \vec{B} \quad \text{where} \quad \vec{A} \perp \vec{B}$$

MAGNITUDES of resultant sum $[\vec{R}_{(+)}]$ and difference $[\vec{R}_{(-)}]$ vectors *when* $\vec{A} \perp \vec{B}$:

Pythagora's Theorem : $R_{(+)} = R_{(-)} = \sqrt{A^2 + B^2}$ or $|\vec{R}_{(\pm)}| = \sqrt{A^2 + B^2}$

Addition of Two Perpendicular Vectors (2D)



Parallelogram Method (Toe-to-Toe / Tail-to-Tail)

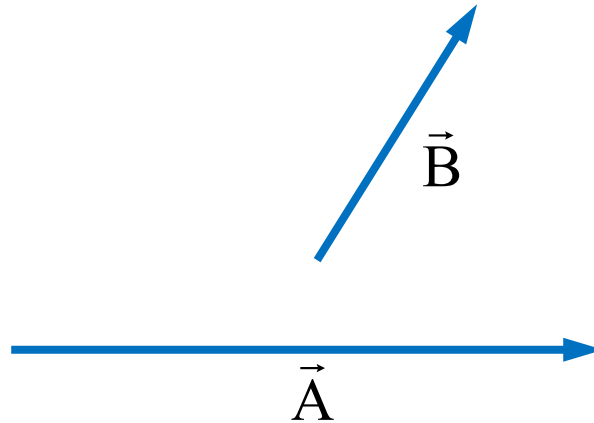
$$\vec{R}_{(+)} = \vec{B} + \vec{A} \text{ and } \vec{R}_{(-)} = (-\vec{B}) + \vec{A} \quad \text{where } \vec{A} \perp \vec{B}$$

MAGNITUDES of resultant sum $[\vec{R}_{(+)}]$ and difference $[\vec{R}_{(-)}]$ vectors *when* $\vec{A} \perp \vec{B}$:

$$\text{Pythagora's Theorem : } R_{(+)} = R_{(-)} = \sqrt{A^2 + B^2} \quad \text{or} \quad |\vec{R}_{(\pm)}| = \sqrt{A^2 + B^2}$$

Addition of Any Two Vectors in Plane (2D)

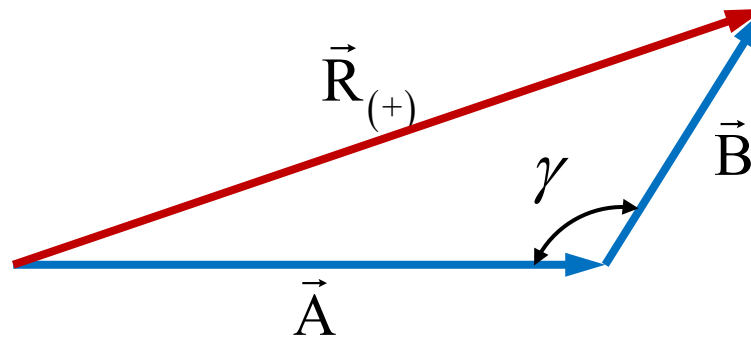
Polygon Rule (Tip - to - Toe or Head - to - Tail)



Translate/slide vector $\vec{B}$ *without* rotating/tilting it until its toe/tail touches the tip/head of vector $\vec{A}$. Then draw the resultant sum vector $\vec{R}_{(+)} = \vec{A} + \vec{B}$ by connecting the *toe* of $\vec{A}$ with the *tip* of $\vec{B}$. This procedure can be continued *ad infinitum* for more than two vectors - see example on board.

Addition of Any Two Vectors in Plane (2D)

Polygon Rule (Tip - to - Toe or Head - to - Tail)



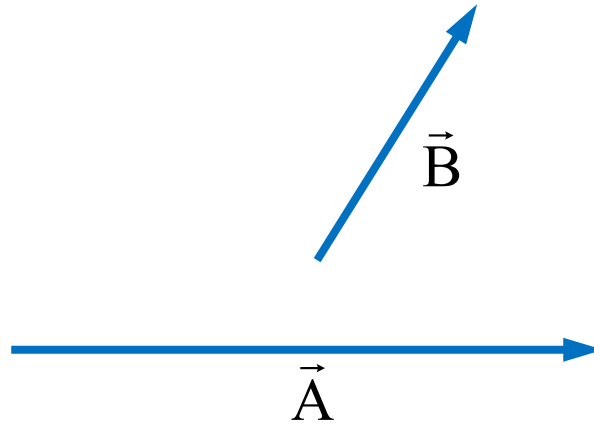
$$\vec{R}_{+} = \vec{A} + \vec{B} = \vec{B} + \vec{A}$$

General Expression for the magnitude of the resultant sum vector $\vec{R}_{(+)}$:

Law of Cosines : $R_{(+)} = \sqrt{A^2 + B^2 - 2 \cdot A \cdot B \cdot \cos(\gamma)}$

Addition of Any Two Vectors in Plane (2D)

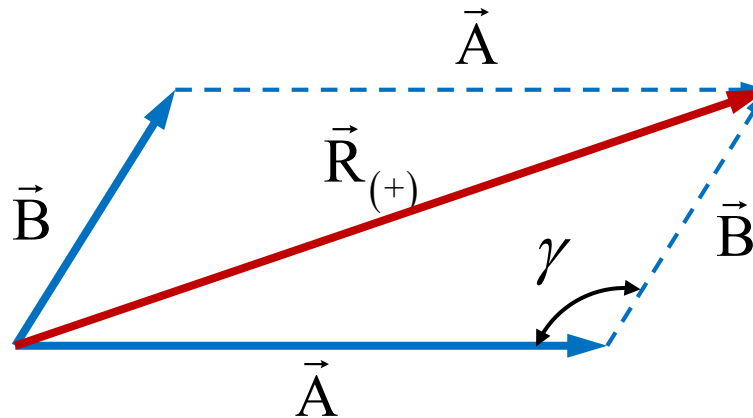
Parallelogram Rule (Toe-to-Toe or Tail-to-Tail)



Translate/slide vector $\vec{B}$ without rotating/tilting it until its toe/tail touches the toe of vector $\vec{A}$. Then draw a parallel to vector $\vec{A}$ starting from the tip/head of $\vec{B}$ and another parallel to $\vec{B}$ starting from the tip of $\vec{A}$. The resultant sum vector $\vec{R}_{(+)} = \vec{A} + \vec{B}$ is obtained by connecting with an arrow the common toe of $\vec{A}$ and $\vec{B}$ with the point of intersection of the two parallels. The toe of $\vec{R}_{(+)}$ should coincide with the common toe of $\vec{A}$ and $\vec{B}$.

Addition of Any Two Vectors in Plane (2D)

Parallelogram Rule (Toe-to-Toe or Tail-to-Tail)



$$\vec{R}_{(+)} = \vec{A} + \vec{B} = \vec{B} + \vec{A}$$

General Expression for the magnitude of the resultant sum vector $\vec{R}_{(+)}$:

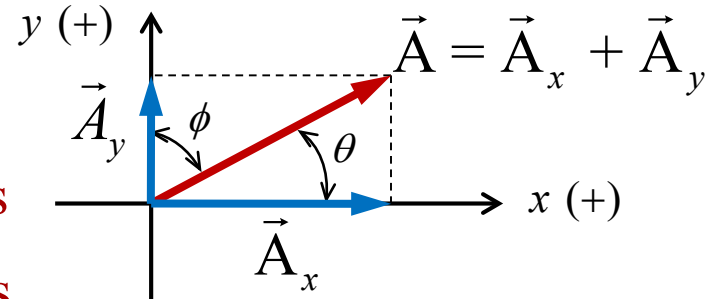
Law of Cosines : $R_{(+)} = \sqrt{A^2 + B^2 - 2 \cdot A \cdot B \cdot \cos(\gamma)}$

(Vector) Components of a Vector (2D)

System of Coordinates: (x, y)

$\vec{A}_x$ is the (vector) component of $\vec{A}$ on the x -axis

$\vec{A}_y$ is the (vector) component of $\vec{A}$ on the y -axis



Knowing A and θ or ϕ we can figure out A_x and A_y :

$$A_x = A \cdot \cos(\theta) = A \cdot \sin(\phi)$$

$$A_y = A \cdot \sin(\theta) = A \cdot \cos(\phi)$$

Knowing A_x and A_y we can

figure out A and θ or ϕ :

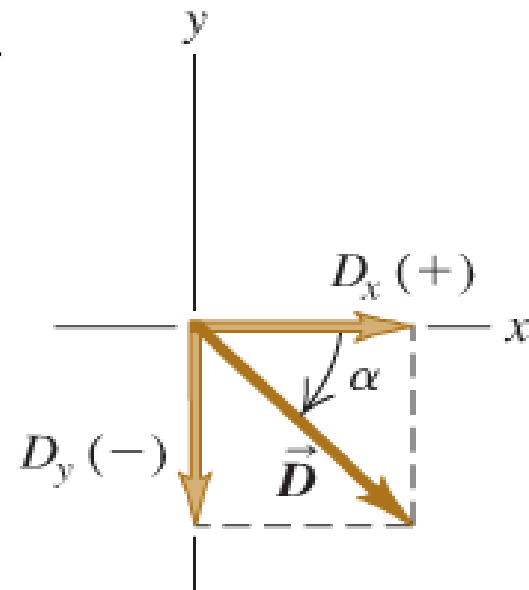
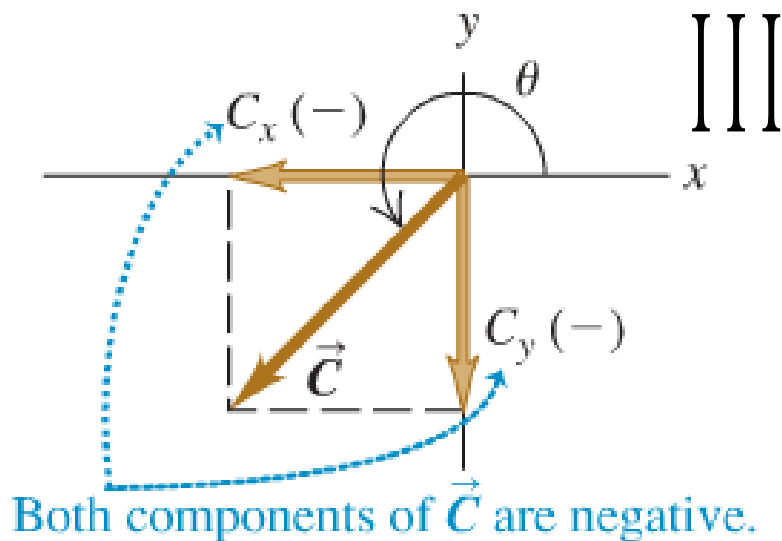
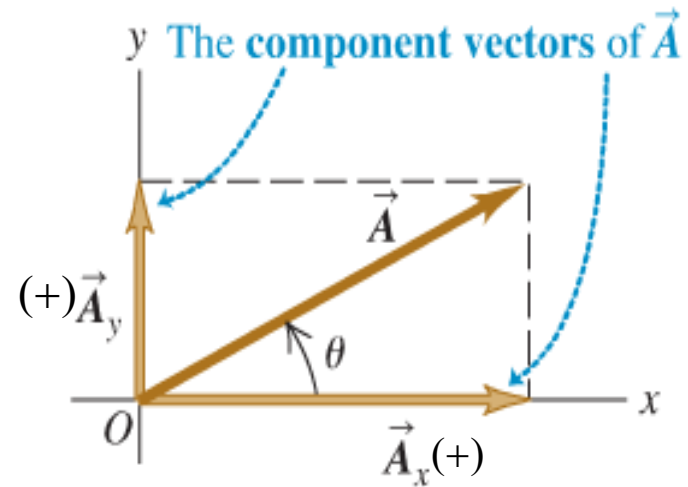
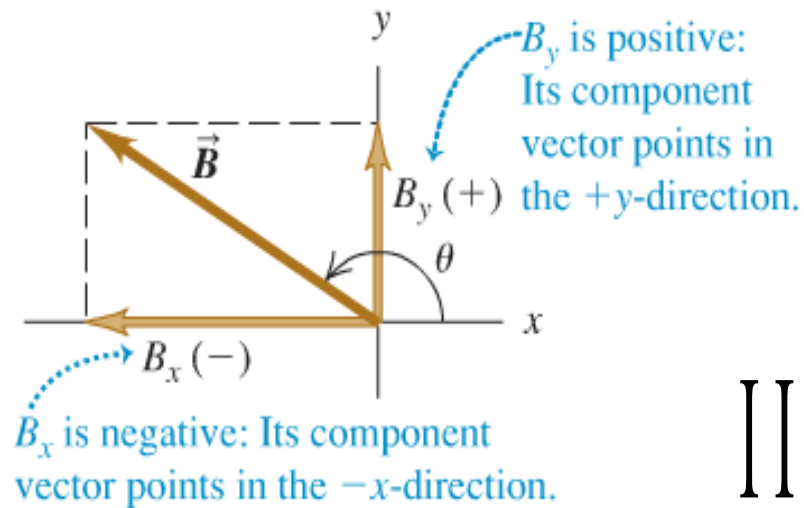
magnitude: $A = \sqrt{A_x^2 + A_y^2}$

$$\tan(\theta) = \frac{A_y}{A_x} \Rightarrow \theta = \arctan\left(\frac{|A_y|}{|A_x|}\right)$$

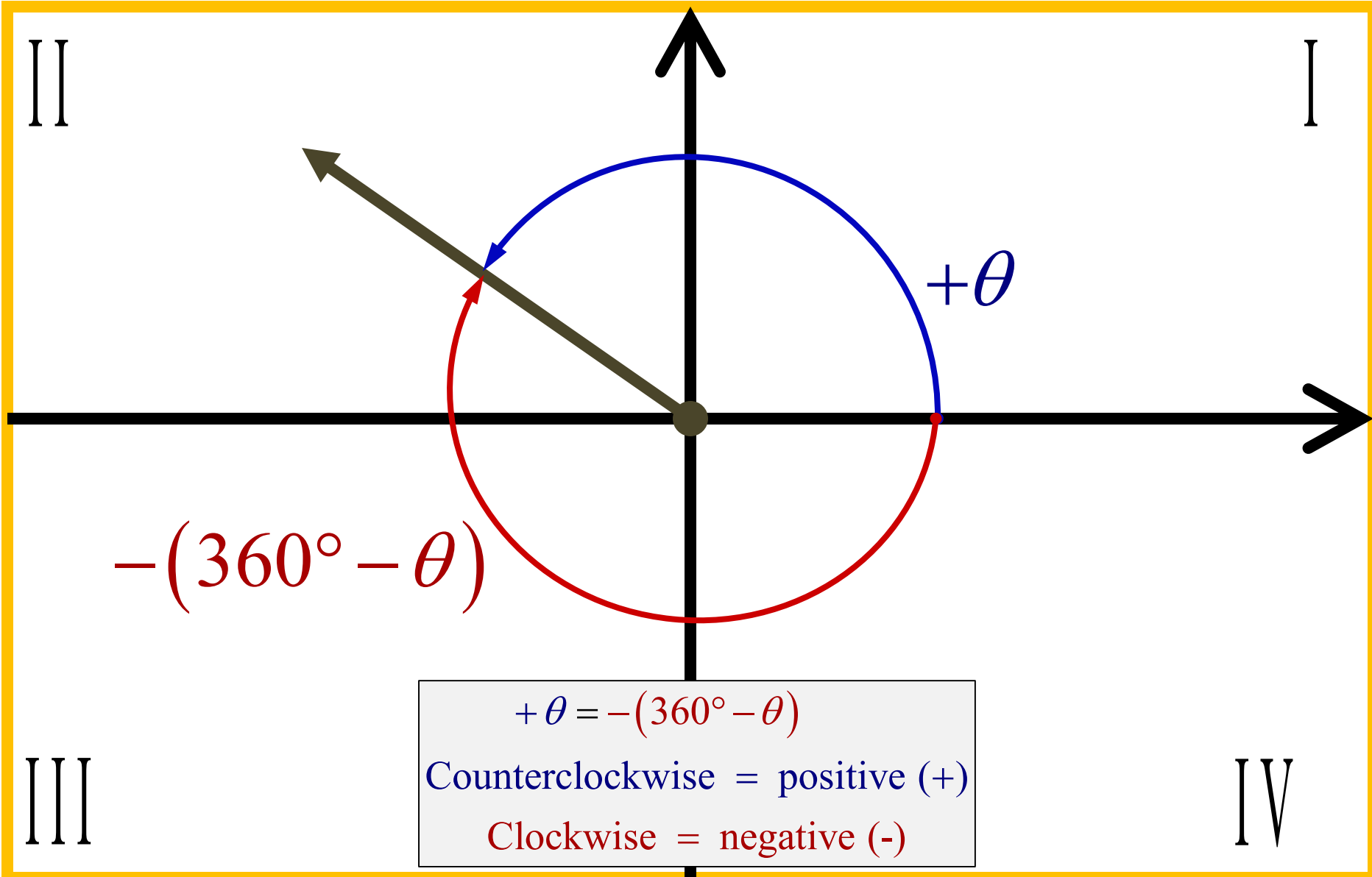
$$\tan(\phi) = \frac{A_x}{A_y} \Rightarrow \phi = \arctan\left(\frac{|A_x|}{|A_y|}\right)$$

But pay attention to the **QUADRANT** in which the particular vector is located!

Quadrants – Illustrative Clarification



Angle Reading – Illustrative Clarification



The Addition of Vectors by Components (2D)

If we have 2 vectors $\vec{A}$ and $\vec{B}$ we can add them all up to generate a resultant vector $\vec{R} = \vec{A} + \vec{B}$ having the *magnitude*:

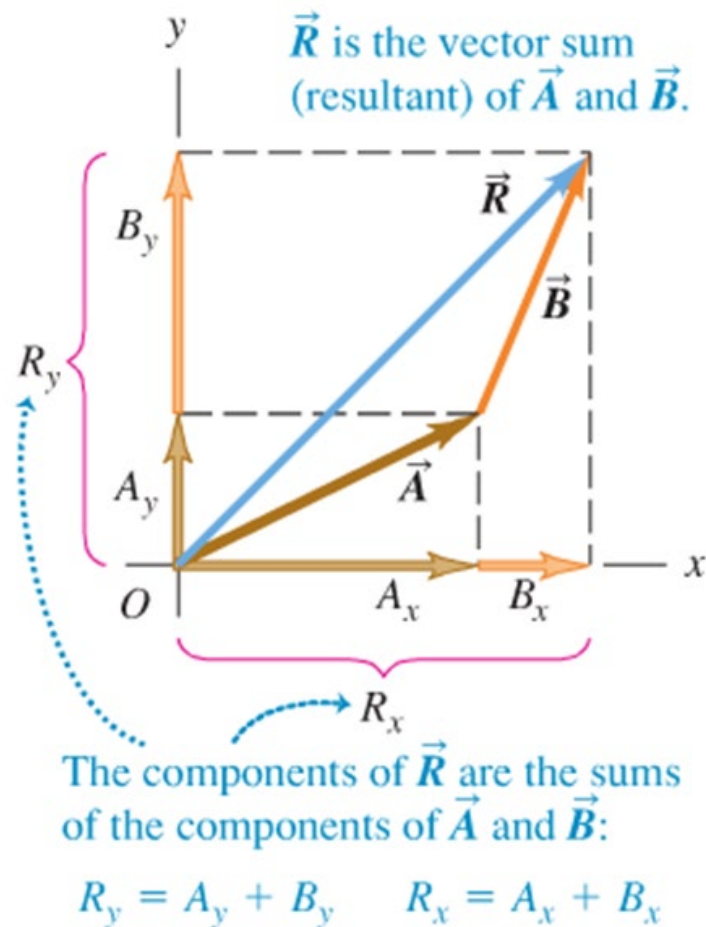
$$R = \sqrt{R_x^2 + R_y^2},$$

with:
$$\begin{cases} R_x = A_x + B_x \\ R_y = A_y + B_y \end{cases}$$

and *angles* θ_R or ϕ_R :

$$\tan(\theta_R) = \frac{|R_y|}{|R_x|}, \text{ which quadrant?}$$

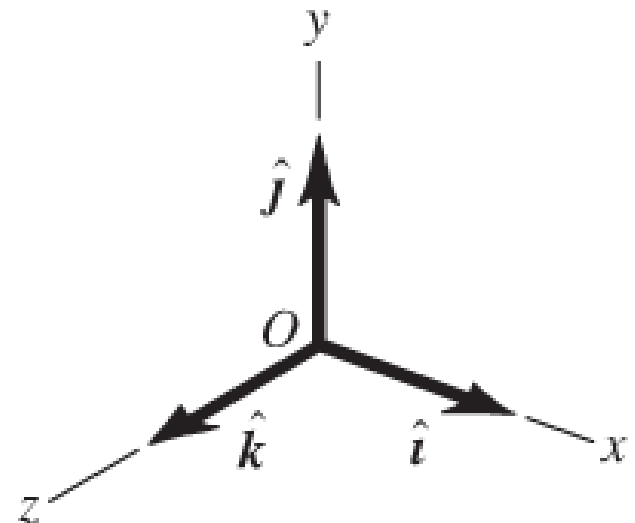
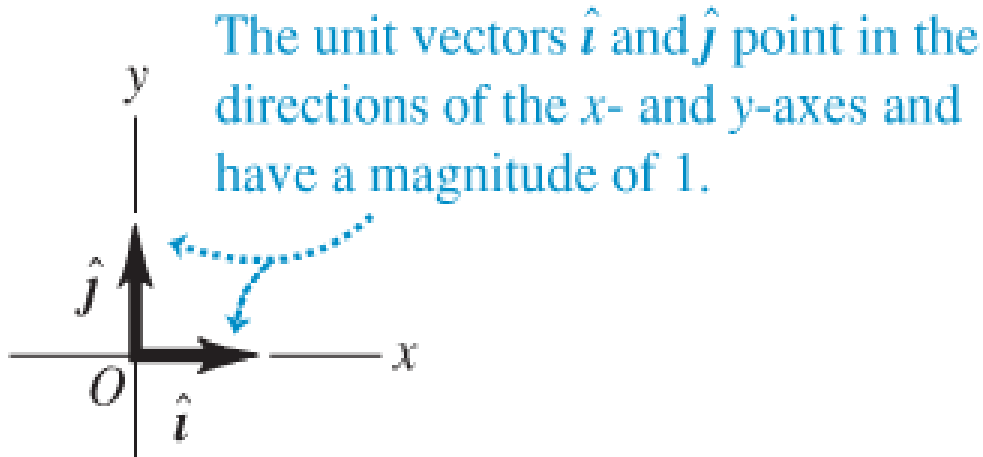
$$\tan(\phi_R) = \frac{|R_x|}{|R_y|}, \text{ which quadrant?}$$



Unit Vectors

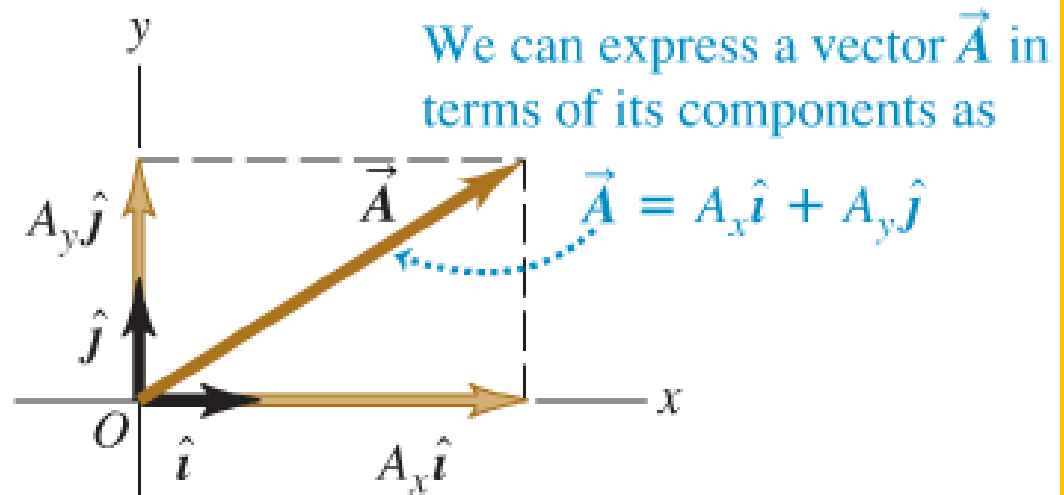
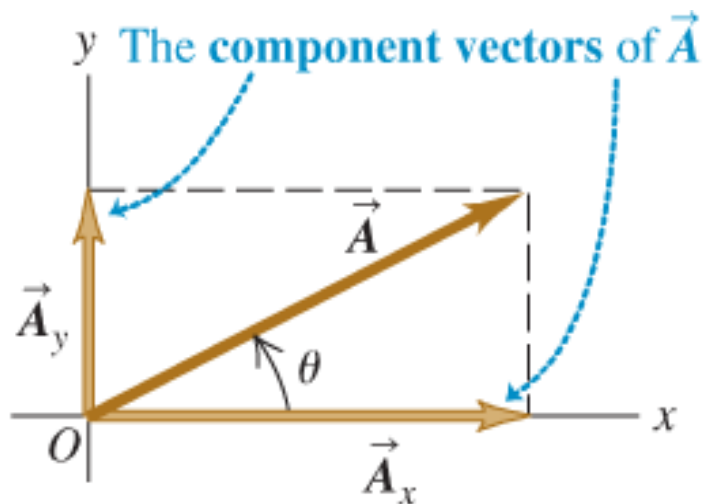
The UNIT VECTOR is a vector of *magnitude equal to 1 and without units*; its only *purpose* is to point towards a direction in space - the positive directions of x , y , and z axes.

The unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$.



Comparison of Vector Representations

Here we present a Comparison between the Vector Components representation and the Unit-Vectors one.



Components Representation:

$$\vec{A} = \vec{A}_x + \vec{A}_y$$

Unit-Vectors Representation:

$$\vec{A} = A_x \cdot \hat{i} + A_y \cdot \hat{j}$$

Addition/Subtraction of Vectors in Space (3D)

Components Format :

$$\vec{A} = \vec{A}_x + \vec{A}_y + \vec{A}_z$$

$$\vec{B} = \vec{B}_x + \vec{B}_y + \vec{B}_z$$

Unit Vectors Format :

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{C} = \vec{A} \pm \vec{B} = (\vec{A}_x \pm \vec{B}_x) + (\vec{A}_y \pm \vec{B}_y) + (\vec{A}_z \pm \vec{B}_z) = \vec{C}_x + \vec{C}_y + \vec{C}_z$$

$$\vec{C} = \vec{A} \pm \vec{B} = (A_x \pm B_x) \hat{i} + (A_y \pm B_y) \hat{j} + (A_z \pm B_z) \hat{k} = C_x \hat{i} + C_y \hat{j} + C_z \hat{k}$$

$$C = \sqrt{C_x^2 + C_y^2 + C_z^2}, \text{ with: } C_x = A_x \pm B_x, C_y = A_y \pm B_y, C_z = A_z \pm B_z$$

Vector Addition Using Unit Vectors – Example #1

Determine $\vec{C}(C_x, C_y, C_z)$ such that $\vec{C} = \vec{A} + \vec{B}$.

Where $\vec{A}(0, -3, 2)$ and $\vec{B}(-4, 3, -1)$.

$$\vec{A} = 0\hat{i} - 3\hat{j} + 2\hat{k} \quad \text{and} \quad \vec{B} = -4\hat{i} + 3\hat{j} - 1\hat{k}$$

$$\vec{C} = (0 - 4)\hat{i} + (-3 + 3)\hat{j} + (2 - 1)\hat{k} = -4\hat{i} + \hat{k}$$

$$\text{So: } \vec{C}(-4, 0, 1).$$

The Unit Vector Corresponding to a Vector

One can turn **any (nonzero) vector** $\vec{u}$ into its corresponding unit vector $\hat{u}$ by simply dividing the vector over its own magnitude $|\vec{u}|$.

$$\hat{u} = \frac{\vec{u}}{|\vec{u}|} = \frac{u_x \hat{i} + u_y \hat{j} + u_z \hat{k}}{\sqrt{u_x^2 + u_y^2 + u_z^2}} = \left(\frac{u_x}{\sqrt{u_x^2 + u_y^2 + u_z^2}} \right) \hat{i} + \left(\frac{u_y}{\sqrt{u_x^2 + u_y^2 + u_z^2}} \right) \hat{j} + \left(\frac{u_z}{\sqrt{u_x^2 + u_y^2 + u_z^2}} \right) \hat{k}$$

Example #2 : Figure out the unit vector for $\vec{w} = 3\hat{i} - 2\hat{j} + \hat{k}$.

$$\hat{w} = \frac{3\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{3^2 + (-2)^2 + 1^2}} = \left(\frac{3}{\sqrt{14}} \right) \hat{i} + \left(\frac{-2}{\sqrt{14}} \right) \hat{j} + \left(\frac{1}{\sqrt{14}} \right) \hat{k}$$

Let's verify that the magnitude of $\hat{w}$ is actually one (1):

$$|\hat{w}| = \sqrt{\left(\frac{3}{\sqrt{14}} \right)^2 + \left(\frac{-2}{\sqrt{14}} \right)^2 + \left(\frac{1}{\sqrt{14}} \right)^2} = \frac{\sqrt{14}}{\sqrt{14}} = 1.$$

Product Between a Scalar and a Vector

The *product between a scalar and a vector* is a new vector having its new magnitude equal to the absolute value of the scalar multiplied by the magnitude of the old vector.

This new vector is *always* parallel to the old vector and having the same or opposite direction to the old one depending on the sign of the multiplying scalar.

Example #3 : Given $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$, figure out the magnitude of the new vector $\vec{b} = -3\vec{a}$.

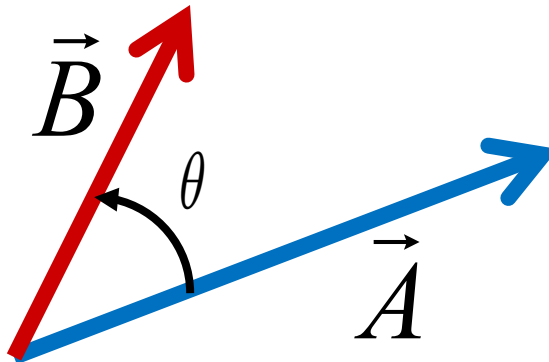
$$|\vec{b}| = |-3\vec{a}| = |-3||\vec{a}| = 3\sqrt{3^2 + 1^2 + (-2)^2} = 3\sqrt{14}.$$

Scalar (Dot) Product of Two Vectors

The Scalar (Dot •) Product of two vectors is a SCALAR.

Definition: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A} = |\vec{A}| |\vec{B}| \cos(\theta) = AB \cos(\theta)$

θ is the angle between the two vectors *after* being aligned *toe-to-toe* (*tail-to-tail*).



Scalar (Dot) Product of Two Vectors

The Scalar Product of two vectors that are *perpendicular* is always ZERO because $\theta = \pm 90^\circ$ and $\cos(\pm 90^\circ) = 0$.

The CONDITION that two (nonzero) vectors be perpendicular is that their Scalar Product be ZERO.

The Scalar Product of Unit Vectors of the *same type* is **1** and of *different type* is **0**.

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = (1)(1) \cos 0^\circ = 1$$

$$\hat{i} \cdot \hat{j} = \hat{i} \cdot \hat{k} = \hat{j} \cdot \hat{k} = (1)(1) \cos 90^\circ = 0$$

Another Expression for Scalar (Dot) Product

Another Important Expression for the Scalar Product of two vectors as a function of their components is:

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Proof : $\vec{A} \cdot \vec{B} = (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \cdot (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) = A_x B_x \underbrace{(\hat{i} \cdot \hat{i})}_1 +$
 $A_y B_x \underbrace{(\hat{j} \cdot \hat{i})}_0 + A_z B_x \underbrace{(\hat{k} \cdot \hat{i})}_0 + A_x B_y \underbrace{(\hat{i} \cdot \hat{j})}_0 + A_y B_y \underbrace{(\hat{j} \cdot \hat{j})}_1 + A_z B_y \underbrace{(\hat{k} \cdot \hat{j})}_0 +$
 $A_x B_z \underbrace{(\hat{i} \cdot \hat{k})}_0 + A_y B_z \underbrace{(\hat{j} \cdot \hat{k})}_0 + A_z B_z \underbrace{(\hat{k} \cdot \hat{k})}_1 = A_x B_x + A_y B_y + A_z B_z.$

Quiz #1

Calculate x such that the two vectors $\vec{U} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{W} = x\hat{i} + 4\hat{j} + 2\hat{k}$ are perpendicular (normal) to each other.

A. $+0.5$

B. -1

C. $+1.5$

D. -2

Quiz #2

Calculate the dot product $\vec{A} \cdot \vec{B}$ between $\vec{A} = 3\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{B} = \hat{i} + 2\hat{j} - 4\hat{k}$.

A. $+5$

B. -5

C. $-5\hat{k}$

D. 0

The Angle Between Two Vectors

To calculate the angle between two vectors $\vec{A}$ and $\vec{B}$ we combine the two expressions for the scalar product: $\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}|\cos(\theta) = A_x B_x + A_y B_y + A_z B_z$

$$\cos(\theta) = \frac{A_x B_x + A_y B_y + A_z B_z}{|\vec{A}||\vec{B}|} = \frac{A_x B_x + A_y B_y + A_z B_z}{\left(\sqrt{A_x^2 + A_y^2 + A_z^2}\right)\left(\sqrt{B_x^2 + B_y^2 + B_z^2}\right)}.$$

Example #4 :

Calculate the angle θ between $\vec{A} = 3\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{B} = \hat{i} + 2\hat{j} - 4\hat{k}$.

$$\cos(\theta) = \frac{(3)(1) + (-2)(2) + (1)(-4)}{\left(\sqrt{3^2 + (-2)^2 + 1^2}\right)\left(\sqrt{1^2 + 2^2 + (-4)^2}\right)} = \frac{-5}{\sqrt{14} \times \sqrt{21}} = \frac{-5}{\sqrt{294}}$$

$$\cos(\theta) = -\frac{5}{17.1464} \Rightarrow \theta = \arccos\left(-\frac{5}{17.1464}\right) \cong 107^\circ (\text{or } -253^\circ).$$

Quiz #3

Calculate the angle θ between $\vec{A} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{B} = 3\hat{i} + 2\hat{j} - \hat{k}$.

A. 312.4° (-47.6°)

B. 205.3° (-154.7°)

C. 180° (-180°)

D. 87.8° (-272.2°)

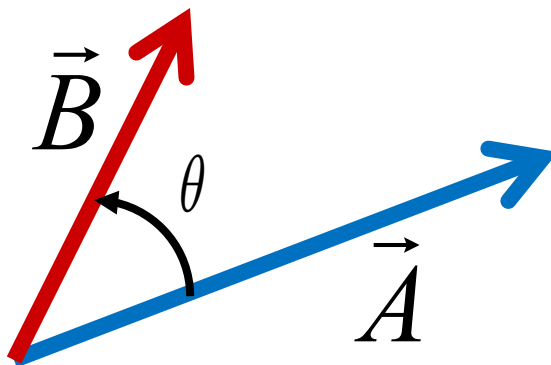
E. 102.6° (-257.4°)

Vector (Cross) Product of Two Vectors

The Vector (Cross $\times$) Product of two vectors is a VECTOR.

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}, \text{ and } |\vec{A} \times \vec{B}| = |\vec{A}| |\vec{B}| \sin(\theta) = AB |\sin(\theta)|$$

θ is the angle between the two vectors *after* being aligned *toe-to-toe* (*tail-to-tail*). The angle starts on $\vec{A}$ and ends on $\vec{B}$ for $\vec{A} \times \vec{B}$ and starts on $\vec{B}$ and ends on $\vec{A}$ for $\vec{B} \times \vec{A}$.



Vector (Cross) Product of Two Vectors

The Vector Product of two vectors that are *parallel* having the *same* or *opposite* direction is always ZERO because θ is either 0° or 180° , respectively and $\sin(0^\circ) = \sin(180^\circ) = 0$.

The CONDITION that two (nonzero) vectors be *parallel* having same or opposite directions is that their **vector product be ZERO**.

The Vector Product of Unit Vectors of the *same type* is always **0**.

$$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$$

$$\hat{i} \times \hat{j} = -\hat{j} \times \hat{i} = \hat{k}$$

$$\hat{j} \times \hat{k} = -\hat{k} \times \hat{j} = \hat{i}$$

$$\hat{k} \times \hat{i} = -\hat{i} \times \hat{k} = \hat{j}$$

Another Expression for Vector (Cross) Product

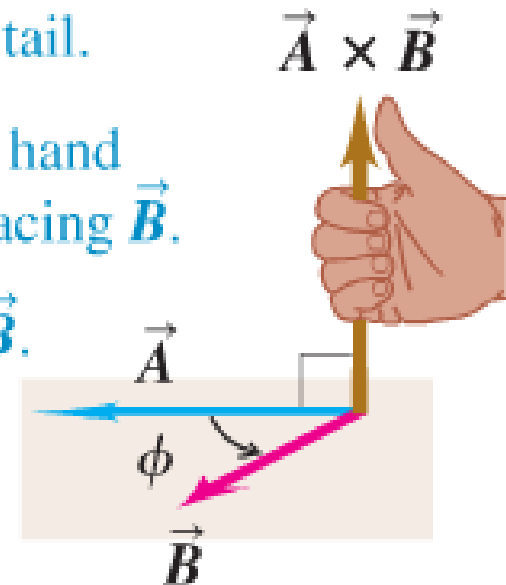
Another **important** expression for the Vector Product of two vectors is:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_y B_z - A_z B_y) \hat{i} + (A_z B_x - A_x B_z) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$$

Determining the direction of new vector $\vec{A} \times \vec{B}$ – using the *Right Hand Rule*.

positive $\equiv$ counterclockwise
negative $\equiv$ clockwise

- ① Place $\vec{A}$ and $\vec{B}$ tail to tail.
- ② Point fingers of right hand along $\vec{A}$, with palm facing $\vec{B}$.
- ③ Curl fingers toward $\vec{B}$.
- ④ Thumb points in direction of $\vec{A} \times \vec{B}$.



Quiz #4

Calculate the cross product $\vec{A} \times \vec{B}$ between $\vec{A} = 3\hat{i} - 2\hat{k}$ and $\vec{B} = -\hat{i} + 4\hat{j}$.

A. $4\hat{i} + \hat{j} - 6\hat{k}$

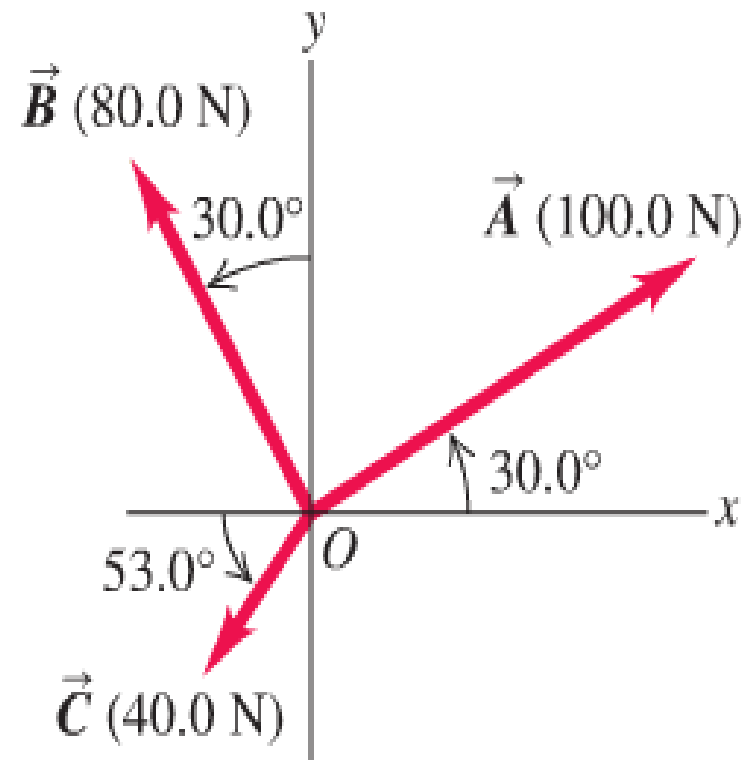
B. $8\hat{i} - \hat{j} + 6\hat{k}$

C. $6\hat{i} + \hat{j} + 3\hat{k}$

D. $2(4\hat{i} + \hat{j} + 6\hat{k})$

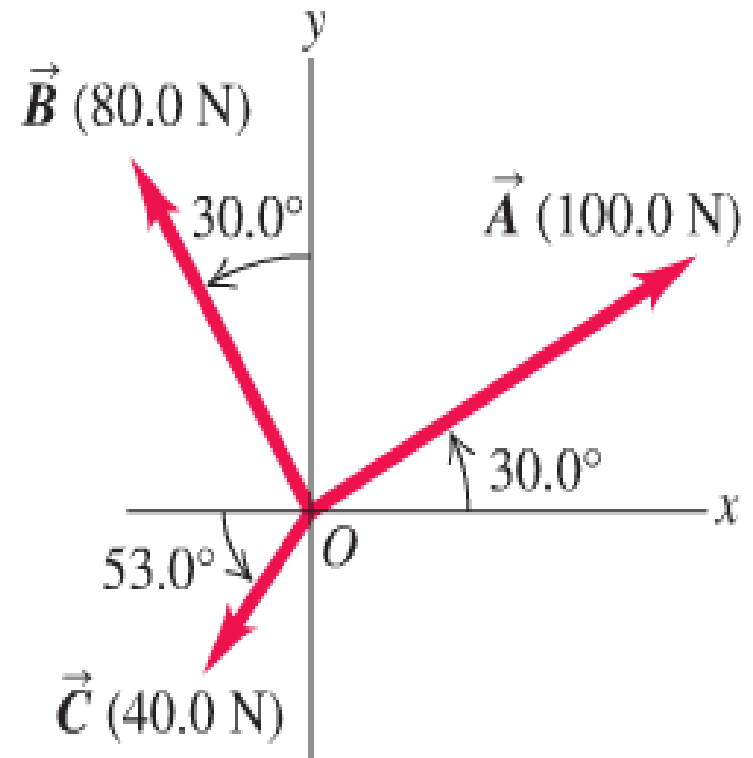
(Extra) Example #5

Calculate the components of all three vectors in the picture.



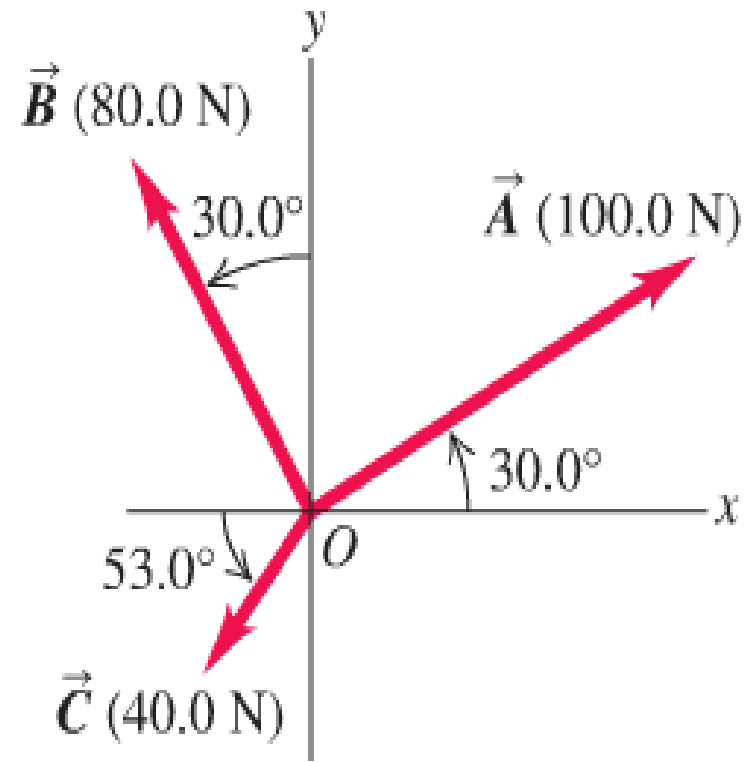
(Extra) Example #6

Calculate the unit vectors $\hat{a}$, $\hat{b}$ and $\hat{c}$ corresponding to $\vec{A}$, $\vec{B}$ and $\vec{C}$, respectively.



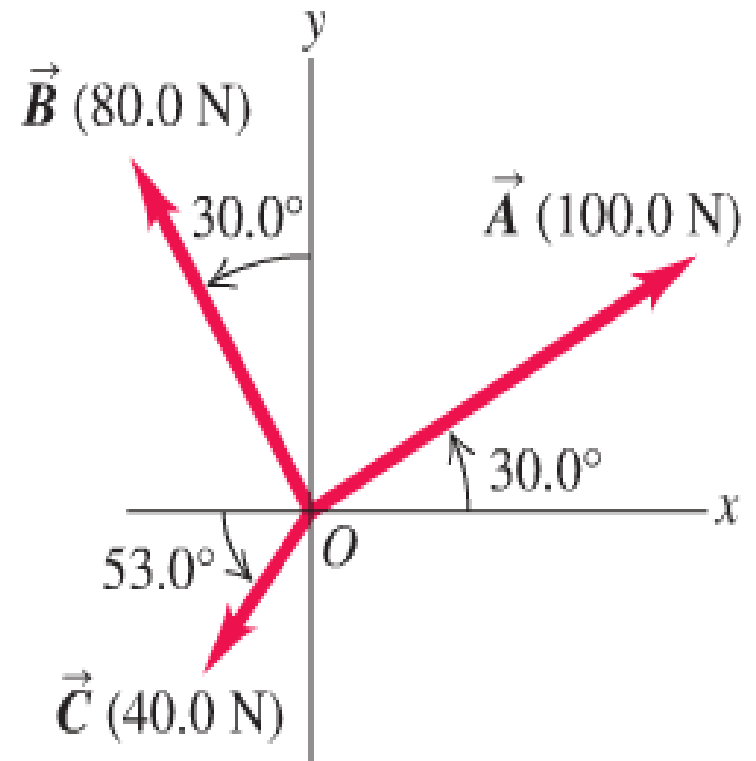
(Extra) Example #7

Calculate the scalar (dot or inner) product $\vec{A} \cdot \vec{B}$.



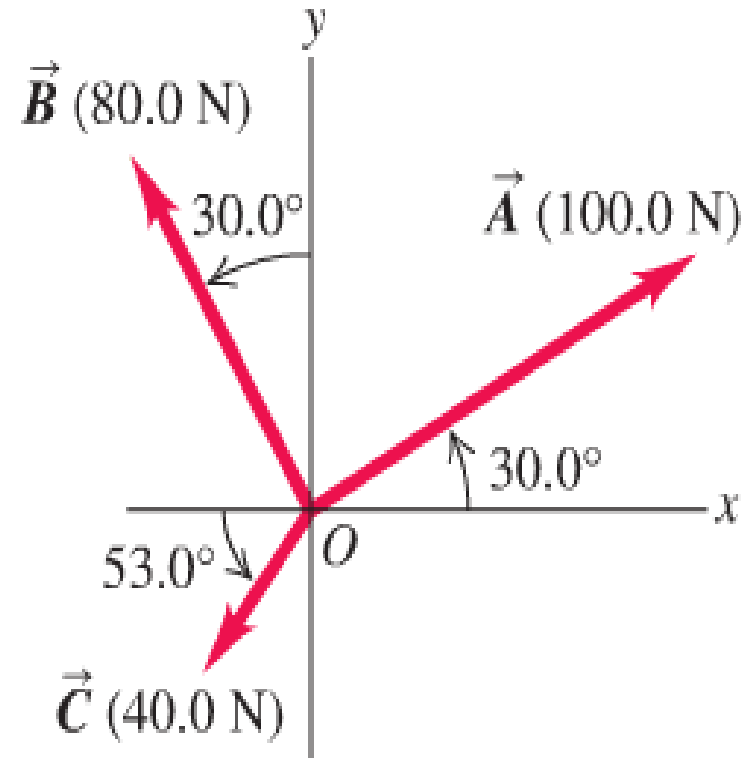
(Extra) Example #8

Calculate the angle between $\vec{A}$ and $\vec{B}$.



(Extra) Example #9

Calculate the vector (cross) product $\vec{A} \times \vec{B}$.



(Extra) Quiz #5

Calculate the unit vector $\hat{\mathbf{b}}$ corresponding to $\vec{B} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 5\hat{\mathbf{k}}$.

A. $\hat{\mathbf{b}} = \left(\frac{1}{\sqrt{30}}\right)\hat{\mathbf{i}} + \left(\frac{\sqrt{2}}{\sqrt{6}}\right)\hat{\mathbf{j}} + \left(\frac{-\sqrt{6}}{\sqrt{5}}\right)\hat{\mathbf{k}}$

B. $\hat{\mathbf{b}} = \left(\frac{1}{\sqrt{30}}\right)\hat{\mathbf{i}} + \left(\frac{1}{\sqrt{15}}\right)\hat{\mathbf{j}} - \left(\frac{1}{\sqrt{5}}\right)\hat{\mathbf{k}}$

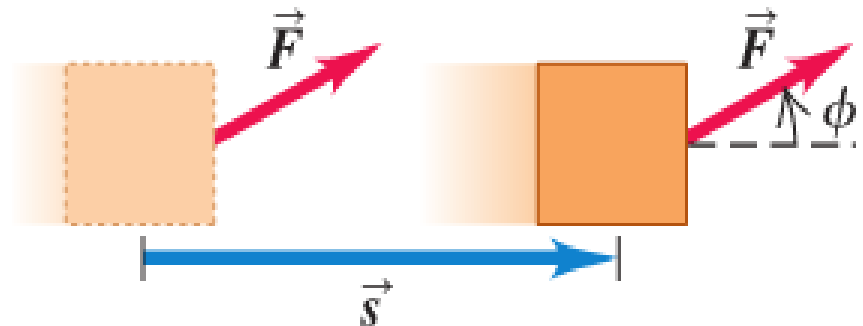
C. $\hat{\mathbf{b}} = \left(\frac{1}{\sqrt{30}}\right)\hat{\mathbf{i}} + \left(\frac{\sqrt{2}}{\sqrt{15}}\right)\hat{\mathbf{j}} - \left(\frac{\sqrt{5}}{\sqrt{6}}\right)\hat{\mathbf{k}}$

(Extra) Quiz #6

We shall learn later that the (mechanical) Work is the *scalar product* between the (vector) Displacement ($\vec{s}$) of an object and the (vector) Force ($\vec{F}$) acting on that object ($W = \vec{F} \cdot \vec{s}$).

Which component of that force $\vec{F}$ actually does the Work ?

- A. *Only* the component of $\vec{F}$ that is *perpendicular* to $\vec{s}$.
- B. *Only* the component of $\vec{F}$ that is *parallel* to $\vec{s}$.
- C. *Both* components of $\vec{F}$.
- D. *None* of the above.



(Extra) Quiz #7

We shall learn later that the (vector) Torque is given by: $\vec{\tau} = \vec{r} \times \vec{F}$.
Which component of that force $\vec{F}$ actually generates the Torque ($\vec{\tau}$),
 $\vec{F}_{\text{tan}}$ or $\vec{F}_{\text{rad}}$? **NOTE:** Neglect most of those fancy writings in the picture below.

- A. *Only* the one *perpendicular* to $\vec{r}$ ($\vec{F}_{\text{tan}}$)
- B. *Only* the one *parallel* to $\vec{r}$ ($\vec{F}_{\text{rad}}$)
- C. *Both* components of $\vec{F}$ ($\vec{F}_{\text{tan}}$ and $\vec{F}_{\text{rad}}$)
- D. *None* of the above

